

Sadif Ahmed

☎ (880)1521564856 ✉ ahmedsadif67@gmail.com ✉ sadif.ahmed@bracu.ac.bd in [in/sadif-ahmed-092b60307](https://www.linkedin.com/in/sadif-ahmed-092b60307)
🌐 [Sadif-Ahmed](https://www.sadif-ahmed.com) 🌐 [sadif-ahmed.github.io](https://github.com/sadif-ahmed) 📍 Dhaka, Bangladesh

ABOUT ME

I'm a passionate and versatile researcher with a keen interest in software engineering and LLMs. I enjoy the art of teaching.

RESEARCH INTERESTS

Software Engineering and Security, Web UI Automation, AI for Software Engineering.

EDUCATION

Bangladesh University of Engineering & Technology Dhaka, Bangladesh
B.Sc. Computer Science, *CGPA: 3.83 out of 4 [Final Year: 3.96/4.00]* Feb 2020 - March 2025
Related Coursework: Structured Programming Language (C), Data Structures and Algorithms (C++), Object Oriented Programming (C++, Java), Software Engineering (Java), Information Systems Design, Database (SQL), Numerical Methods (Python), Computer Architecture, Machine Learning (Python)
Notre Dame College Dhaka, Bangladesh
Higher Secondary School Certificate, *GPA: 5 out of 5* Jan 2017 - Jan 2019
St. Joseph High School Dhaka, Bangladesh
Secondary School Certificate, *GPA: 5 out of 5* Jan 2009 - Dec 2016

PROFESSIONAL EXPERIENCE

- **Lecturer, CSE Department, BRAC University, Dhaka, Bangladesh** July 2025 - Present
Courses Taught:
 - Computer Architecture
 - Software Engineering
- **Research Assistant, CSE Department, BUET, Dhaka, Bangladesh** March 2025 - Present
 - Explored WebUI Gyms such as **WebArena** and worked on a UI Testing Automation Pipeline.
 - Worked on creating a end to end pipeline to generate **playwright** testing scripts from user stories of a website.
 - Building and fine-tuning a bilingual (Bangla/English) **vision-language model** optimized for Bengali language generation and understanding, incorporating curriculum-aligned pedagogy such as **Socratic prompts** and step-by-step scaffolding.
 - Implementing personalized **student modeling** features, including knowledge tracing and bilingual hinting, to dynamically adapt item difficulty based on user performance.

RESEARCH EXPERIENCE

- **Secret Breach Detection in Source Code with Large Language Models**
Undergraduate Thesis, **ESEM 2025 Technical Track Publication** October 2024 - July 2025
 - **Key Contribution:** We introduce a novel approach for Secret Breach Detection in source code using a Small Language Model (SLM) fine-tuned with QLoRA. Our model demonstrably outperforms several established state-of-the-art regex-based tools (like Trufflehog) and large, general-purpose LLMs (like GPT-4o) on the SecretBench dataset. We establish the efficacy of leveraging compact, specialized models over large, zero-shot models for this specific, critical software security task.
 - **Technology & Tools:** QLoRA, DeepSeek-7B, Gemma-7B, LLaMA-3.1-8B, Mistral-7B, DeepSeek-V3, GPT-4o
 - **Supervisor:** Dr. Rifat Shahriyar, Professor, CSE, BUET; Dr. S M Sohan, Sr. Engineering manager, Google LLC
- **Secret Leak Detection in Software Issue Reports using LLMs: A Comprehensive Evaluation**
Undergraduate Thesis, **MSR 2026 Technical Track Publication, ArXiv** July 2024 - October 2025
 - **Key Contribution:** We present the first large-scale study and a robust hybrid detection pipeline for secret leaks in GitHub issue reports. Our pipeline integrates regex-based extraction with LLM contextual classification to effectively reduce false positives. We curated and released the first public benchmark dataset of over 54,000 labeled instances and demonstrated that fine-tuned LLMs achieve state-of-the-art performance (up to 0.945 F1), significantly outperforming traditional methods.

- **Technology & Tools:** Small Language Models such as RoBERTa-base, BERT-base-cased and BERT-base-uncased, CodeBERT-base etc, GPT-4o, Gemini-2.0-Flash, QLoRA, PEFT, DeepSeek-7B, Gemma-7B, LLaMA-3.1-8B, Mistral-7B, Qwen-7B
 - **Supervisor:** [Dr. Rifat Shahriyar, Professor, CSE, BUET](#); [Dr. Gias Uddin, Associate Professor, York University](#)
- **A Survey on Agentic Security: Applications, Threats and Defenses**
Independent Research Group, [ArXiv](#) August 2025 - October 2025
- **Key Contribution:** We present the first holistic survey of the rapidly evolving agentic security landscape, systematically analyzing over 150 papers published primarily between 2024-2025. We structure the field around three interdependent pillars: Applications, Threats, and Defenses, providing a unified framework to understand the capabilities and vulnerabilities of Large Language Model (LLM) agents in cybersecurity.
 - **Supervisor:** [Dr. Farig Sadeque, Associate Professor, BRAC University](#); [Dr. Md Rizwan Parvez, Scientist, QCRI](#)
- **BanglaForge: LLM Collaboration with Self-Refinement for Bangla Code Generation**
Independent Research Group, Accepted in BLP Workshop co-located with **AACL-IJCNLP 2025** [ACL Anthology](#) August 2025 - September 2025
- **Key Contribution :** We introduce BanglaForge, a novel framework for generating executable code from Bangla descriptions, a low-resource language. We utilize a retrieval-augmented dual-model collaboration paradigm with iterative self-refinement based on execution feedback. This system, combining LLM-based translation and in-context learning, achieves a competitive Pass@1 accuracy of 84.00% on the BLP-2025 Bangla Code Generation benchmark, validating our approach for low-resource code generation.
 - **Technology & Tools:** Dual-LLM architecture, Retrieval-Augmented few-shot prompting, TF-IDF, Iterative self-refinement, Execution feedback, Lg Exaone Deep 32B, Gemini-2.5-Pro
 - **Collaborators :** [Mahir Labib Dihan, Lecturer BRAC University](#); [Md Nafiu Rahman, Lecturer BRAC University](#)
- **ISSUEGUARD: Real-Time Secret Leak Prevention Tool for GitHub Issue Reports**
Undergraduate Thesis, **FSE 2026 Tool Demo Track Publication** [ArXiv](#) October 2025 - February 2026
- **Key Contribution:** Developed a browser-based security tool that provides real-time feedback to developers to prevent accidental secret leaks. The system utilizes a hybrid detection engine (Regex + Fine-tuned CodeBERT) to filter false positives effectively (92.70% F1-score). Engineered a high-performance backend with asynchronous processing and caching to achieve an end-to-end latency of less than 200ms. Conducted a comprehensive user study with 50 developers, where 80% of participants reported "Very High Confidence" in the tool's detection capabilities.
 - **Technology & Tools:** Google Chrome Extension (Manifest V3), JavaScript, FastAPI, Python, CodeBERT-base, PyTorch, FP16 Mixed Precision Inference, LRU Caching.
 - **Supervisor:** [Dr. Rifat Shahriyar, Professor, CSE, BUET](#); [Dr. Gias Uddin, Associate Professor, York University](#)

TECHNICAL SKILLS

- **Languages:** C, C++, Java, Python, JavaScript, TypeScript, LaTeX
- **Frameworks/Libraries:** NodeJS, ExpressJS, SvelteJS, TensorFlow, PyTorch, Matplotlib, NumPy, Pandas
- **Databases:** PostgreSQL, Oracle
- **Tools/Platforms:** Git, GitHub, Linux, Vercel, Supabase, Azure, Velociraptor

ACHIEVEMENTS

Top 20 finalists of Robi Datathon 2024, A countrywide Deep Learning Competition
Dean's list award and university merit scholarship recipient in two terms of undergraduate study in BUET
TalentPool Scholarship for outstanding academic result in the Higher Secondary Certificate Exam in Dhaka, Bangladesh

- **Network Flow Classification and Anomaly Detection**

Technology & Tools: *Python, Pytorch, Tensorflow*

 [Mohaimin41/ml_project](#)

- Developed a **novel pipeline** using **BERT** and **GPT** for **binary and multi-label classification** of anomalous network traffic from pcap data.

- **Machine Learning Algorithms and Neural Network from Scratch**

Technology & Tools: *Python, Numpy, Scikit-learn*

 [Sadif-Ahmed/CSE-472](#)

- Implemented core ML algorithms: **logistic regression** (with bagging/stacking), **SVD** (for image reconstruction), **PCA**, and **GMM/EM** clustering.
- Built a **feed-forward neural network** and the **Adam** optimizer from scratch using only **numpy**.

- **AuthentiDocs - Team Collaboration Authenticated By Digital Signature**

Technology & Tools: *JS, TS, Svelte, PostgreSQL, Supabase*

 [AuthentiDocs/authentidocs](#)

- Created a full-stack document management application integrating file flow, sharing, **digital signature** and verification. Focused on back-end development.

- **Cryptography, Malware Analysis, and Security Attacks**

Technology & Tools: *Python, Docker, Wireshark, Azure*

 [Sadif-Ahmed/CSE-406](#)

- Implemented **AES**, **Diffie-Hellman**, and **RSA** with socket communication. Demonstrated a **buffer-overflow attack** and pedagogical **malware functionalities** in Docker.

- **VLAN Configuration and Wireless Network Simulation**

Technology & Tools: *Java, Cisco Packet Tracer, NS3*

 [Sadif-Ahmed/CSE-322](#)

- Implemented **threaded server-client sockets**. Configured **NAT** and **ACLs** on **VLANs**. Simulated various wired and wireless mobile networks.

- **Operating System Internals with xv6**

Technology & Tools: *Bash, C, Assembly*

 [Sadif-Ahmed/CSE-314](#)

- Explored **bash scripting** and **synchronization** (pthreads). Implemented **system calls** and the **round robin scheduler** in the xv6 operating system.

- **Anidex - Simple Online Anime Database**

Technology & Tools: *JS, Svelte, ExpressJS, PostgreSQL*

 [KyojinsAnidex/Anidex](#)

- Developed a full-stack anime database with features for listing, discovery, and **forum discussions**. Focused on front-end and full-stack integration.

- **Online Utility and Handyman Services**

Technology & Tools: *JS, Svelte, ExpressJS, PostgreSQL*

 [Siam11651/cse326-project](#)

- Full-stack development of a service application using **Svelte** and **PostgreSQL** following a modular design and web development best practices.

REFERENCES

- [Dr. Rifat Shahriyar](#)

Professor, CSE, BUET

✉ rifat.shahriyar@gmail.com, rifat@cse.buet.ac.bd

- [Dr. Md Rizwan Parvez](#)

Scientist, QCRI

✉ rizwan@ucla.edu, rizwan.incipient@gmail.com

- [Dr. Farig Sadeque](#)

Associate Professor, BRAC University

✉ farig.sadeque@bracu.ac.bd

- [Dr. S M Sohan](#)

Sr. Engineering manager, Google LLC

✉ sohansm@google.com